

SEMENIVKA, Ukraine

WMO: 330490

Lat: 52.1908N

Lon: 32.5739E

Elev: 161

StdP: 99.41

Time Zone: 2.00 (EUE)

Period: 86-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.0	(c) -17.1	(d) -23.3	(e) 0.5	(f) -19.7	(g) -20.2	(h) 0.6	(i) -16.7	(j) 12.1	(k) -1.9	(l) 10.5	(m) -2.1	(n) 2.7	(o) 20	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 30.0	(d) 19.4	(e) 28.0	(f) 18.9	(g) 26.3	(h) 18.2	(i) 20.8	(j) 26.9	(k) 19.9	(l) 25.9	(m) 19.0	(n) 25.2	(o) 2.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.0	23.1	17.7	12.9	22.2	16.8	12.2	21.4	60.8	27.3	57.6	25.8	54.7	25.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.7	(b) 7.3	(c) 6.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 8.7	(b) 7.3	(c) 6.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-5.4	-3.7	0.7	7.8	14.2	17.1	18.5	17.9	12.8	6.0	1.1	-3.4
	DBStd	9.80	6.14	5.99	5.33	4.37	4.07	3.20	2.79	3.74	4.06	4.41	5.35	5.26
	HDD10.0	2089	479	383	289	93	8	0	0	0	16	138	269	414
	HDD18.3	4250	737	616	546	315	140	61	33	57	172	382	518	673
	CDD10.0	1003	0	0	1	28	138	213	264	244	100	14	1	0
	CDD18.3	123	0	0	0	0	12	24	38	43	6	0	0	0
	CDH23.3	1338	0	0	0	3	124	202	373	569	66	0	0	0
	CDH26.7	353	0	0	0	0	18	37	79	208	11	0	0	0
Wind	WSAvg	2.8	3.6	3.1	3.4	3.0	2.6	2.4	2.4	2.1	2.3	2.5	2.7	3.4
Precipitation	PrecAvg	780	78	64	60	46	66	77	88	53	58	55	61	74
	PrecMax	1067	167	127	144	102	125	161	152	117	119	117	105	185
	PrecMin	591	28	22	8	3	34	37	28	3	11	5	6	19
	PrecStd	97	30	25	32	22	23	28	35	29	31	32	28	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	9.3	16.8	22.5	28.2	30.1	30.4	33.7	28.2	20.9	13.8	8.3
		MCWB	4.2	6.1	9.8	13.9	17.5	20.3	19.7	19.6	19.2	14.2	12.2	6.2
	2%	DB	3.9	5.3	12.8	20.5	26.5	27.5	28.6	31.2	25.4	17.1	10.7	6.7
		MCWB	2.8	3.6	8.0	12.3	17.7	19.3	19.3	19.3	16.9	13.3	9.4	5.4
	5%	DB	2.8	3.6	9.9	18.1	24.3	25.6	27.0	29.0	22.7	14.6	8.8	4.4
		MCWB	1.7	2.3	5.9	10.8	16.8	18.3	18.9	19.2	15.6	11.9	7.6	3.6
	10%	DB	1.6	2.5	7.5	15.5	21.9	23.7	25.2	26.4	20.1	12.7	6.9	2.2
		MCWB	0.8	1.5	4.6	9.9	15.2	16.8	18.3	17.7	14.7	10.6	5.9	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	6.6	10.0	15.0	20.7	21.9	22.6	21.3	19.5	16.2	12.2	7.1
		MCDB	5.2	8.4	15.5	23.2	25.4	26.8	26.9	29.5	26.7	18.8	13.6	7.6
	2%	WB	3.2	3.9	8.4	13.0	18.7	20.4	20.8	20.1	17.8	13.7	9.8	5.3
		MCDB	3.7	5.2	12.3	18.3	24.5	25.6	26.2	28.2	24.0	16.3	10.4	6.4
	5%	WB	1.9	2.7	6.7	11.9	17.3	19.3	19.9	19.3	16.3	12.2	7.9	3.5
		MCDB	2.6	3.5	9.4	16.7	22.8	23.9	24.9	27.3	20.8	13.9	8.7	4.6
	10%	WB	0.7	1.6	5.0	10.7	16.1	18.0	18.9	18.5	15.0	11.0	6.0	1.5
		MCDB	1.4	2.3	7.2	14.8	20.9	22.6	24.1	25.6	19.7	12.7	6.9	2.0
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	7.2	9.0	10.8	9.8	10.7	12.3	9.5	7.8	4.6	4.3
		MCDBR	3.1	5.5	10.4	13.5	13.4	12.2	13.6	16.5	14.3	11.4	5.5	5.8
		MCWBR	3.0	4.2	6.3	6.8	6.3	5.4	5.8	6.1	7.4	7.0	4.3	5.1
	5% WB	MCDBR	3.2	5.0	9.7	11.8	11.5	10.6	11.5	14.9	12.9	9.1	5.0	5.7
		MCWBR	3.2	4.1	6.2	6.5	5.9	5.3	5.6	6.1	7.2	6.7	4.2	5.1
Clear-Sky Solar Irradiance	taub	0.301	0.334	0.364	0.430	0.402	0.389	0.423	0.427	0.390	0.359	0.338	0.305	
	taud	2.140	2.075	2.137	2.121	2.284	2.347	2.270	2.241	2.304	2.355	2.259	2.181	
	Ebn at Noon	666	750	813	801	850	865	825	797	783	732	624	591	
	Edh at Noon	73	105	121	139	124	118	126	123	103	80	66	60	
All-Sky Solar Radiation	RadAvg	0.81	1.70	3.08	4.15	5.59	5.90	5.62	4.88	3.25	1.69	0.68	0.49	
	RadStd	0.16	0.27	0.47	0.38	0.42	0.50	0.40	0.44	0.38	0.30	0.12	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.65	N/A	N/A	+0.93	+0.56	+0.53	-138	-210	+100	+37

Nomenclature: See separate page